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RESEARCH ARTICLE

Synergy of Polydopamine-Assisted Additive Modification and Hierarchical-Morphology Poly(Vinylidene Fluoride) Nanofiber Mat for Ferroelectric-Assisted Triboelectric Nanogenerator

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Abstract
In the last decade, numerous physical modification methods have been introduced to enhance triboelectric nanogenerator (TENG) performance although they generally require complex and multiple fabrication processes. This study proposes a facile fabrication process for Poly(vinylidene fluoride) (PVDF) nanofiber (NF) mats incorporating additive and nonadditive physical modifications. Patterned PVDF NF mats are prepared by electrospinning using a metal mesh as the NF collector. As a negative triboelectric material, the TENG with the patterned PVDF NF mat exhibits superior performance owing to the engineered morphology of the contact layer. PVDF is crucial in TENGs owing to its superior ferroelectric properties and surface charge density when combined with specific electroceramics. Hence, the synergy of the physical modification methods is achieved by incorporating $BaTiO_3$ (BTO) nanoparticles (NPs) into the PVDF. By functionalizing BTO NPs with polydopamine, the TENG performance is further improved owing to the enhanced dispersion of NPs and improved crystallinity of the PVDF chains. Utilizing large NPs produces a nanopatterning effect on the NF surface, thereby resulting in the hierarchical structure of the NF mats. The source of the voltage signals from the TENG is analyzed using fast Fourier transform.

Keywords Triboelectric nanogenerator · Electrospinning · Physical modification · Poly(vinylidene fluoride) · Barium titanate · Polydopamine

1 Introduction

With the recent surge in societal demand for portable wearable electronics, wireless sensor networks, and Internet of Things (IoT) devices, extensive research has been conducted on harvesting ambient mechanical energy sources to power devices with low energy consumption [1, 2]. Triboelectric nanogenerators (TENGs) [3, 4] and piezoelectric nanogenerators (PENGs) [5, 6] have attracted significant attention as promising methods for harvesting and converting ambient mechanical energy into electrical energy. Triboelectricity arises from two distinct phenomena: contact electrification and electrostatic induction, which occur at the interface of different materials during contact separation motion [3]. Consequently, the power generation efficiency of TENGs can be significantly enhanced through material engineering. Chemical and physical modifications are generally used to enhance the performance of TENGs by increasing the surface charge density of the triboelectric materials [7, 8].

Physical modification methods are classified into additive and nonadditive techniques [8]. Additive physical modifications involve incorporating nanomaterials into the contact layer [9] to increase the surface charge density [10, 11] and prevent the dissipation of triboelectric charges [12, 13]. Numerous studies have demonstrated the performance enhancement in TENGs through the doping of functional fillers, such as high-permittivity nanoparticles (NPs; TiO_2 [14], $SrTiO_3$ [10], and $BaTiO_3$ (BTO) [15]) and graphene oxide [12, 13]. Conversely, nonadditive physical modifications involve altering the contact layer morphology by constructing microstructures and nanostructures to enhance the effective contact area [8]. Micropatterning and nanopatterning of the contact layer using templates are generally

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employed in TENGs to increase their effective contact area [16–18]. The template method offers versatility for creating various structures through mold design modifications. However, it is complex, expensive, and difficult to implement on a large scale. Introducing porous structures into the contact layer of TENGs can partially address the limitations of the template methods as the structures effectively increase the surface area-to-volume ratio and enhance the mechanical, electrical, and thermal properties when applied to TENGs [19]. Porous structures exist in different forms, including foams [20], aerogels [21], hydrogels [22], and fibers [23]. Fibrous structures exhibit a unique morphology that differs from that of other porous structures as they are created from one-dimensional fibers to form fiber mats. Electrospinning, a simple and easy fabrication technique, is crucial in fabricating these fibrous structures. It enables the production of unique nanofiber (NF) morphologies with a high surface area and porosity, thereby enhancing the output performance of TENG [24, 25]. Additionally, electrospun NFs [26, 27] offer significant advantages as the contact layer in TENG when applied to wearable electronics owing to their unique characteristics, such as permeability, wettability, and stretchability [28]. The electrospinning of ferroelectric polymers, such as poly(vinylidene fluoride) (PVDF), and its copolymers can enhance their polar phase content through uniaxial stretching and electrical poling, thereby promoting triboelectric charge transfer [29].

Different methodologies have been employed to increase the effective contact areas of NF-based contact layers by establishing hierarchical structures. Wang et al. [30] developed hierarchically structured hydroxypropyl cellulose/thermoplastic polyurethane NF mats using water-assisted electrospinning followed by drying. Jang et al. [31] demonstrated the formation of honeycomb-like NF networks as the contact layer of a TENG by inducing the self-assembly of poly(vinylidene fluoride–trifluoroethylene) (P(VDF–TrFE)) NFs. An effective hierarchical structure can be generated using numerous processes. Chen et al. [32] introduced a novel strategy for fabricating NF-based hierarchical structures using P(VDF–TrFE) with SiO_2 NPs for electrospinning and prestretched elastomers. However, existing studies faced challenges, such as limited morphology control and complex fabrication processes, thereby requiring further post-processing. Utilizing an electroconductive metal wire mesh as an NF collector offers a potential solution to the aforementioned problems. Owing to the unique hierarchical morphologies of the NF mats, metal wire meshes have been widely used as NF collectors in electrospinning in various fields, such as tissue engineering [33], oil–water separation [34], pressure sensors [35], and air filters [36]. Cui et al. presented a novel type of TENG based on thermoplastic polyurethane NF mat with microhump arrays using a metal wire mesh [37]. Further, they demonstrated a facile method for producing NF-based hierarchical structures with controllable morphologies by varying the mesh sizes although it has limitations. The morphological NF patterns generated by the wire mesh reduced as the fineness of the mesh increased. Further, the morphological patterns faded with an increase in the electrospinning time, thereby leading to a constrained control of the NF mat thickness.

In this study, a patterned PVDF/BTO NF mat is fabricated using an electroconductive wire mesh collector for application in TENGs. The fabrication process enables the morphology control using the mesh size and maintains the patterns on the NF mats, regardless of the mat thickness. Further, the proposed method ensures the integration of additive modifications in the current nonadditive modification process. Herein, BTO NPs are incorporated into a PVDF matrix owing to its outstanding piezoelectric constant and permittivity [38]. They function as nucleating agents, thereby increasing the β-phase content of PVDF [39]. The β-phase of PVDF provides a stronger electrostatic attraction to electrons, thereby enhancing the charge transfer between the contact layers [29]. The performance of TENGs is enhanced by improving the dispersion of the NPs in the PVDF solution using polydopamine (PDA). This study is the first to incorporate PDA-functionalized BTO NPs into a PVDF matrix for application to TENG devices. Additionally, the effect of the BTO NP size on the TENG performance was investigated. Notably, larger NPs induced a nanopatterning effect on the NF surface, thereby resulting in a hierarchical structure within the NF mat. The contributions of the triboelectric and piezoelectric effects on the output voltage signals of the TENG are investigated using signal processing and frequency domain signal analyses.

2 Experimental Section

2.1 Materials

PVDF powder (molecular weight ≈ 534,000 g/mol) was provided by Sigma Aldrich (St. Louis, MO, USA). Acetone and dimethylformamide (DMF) solutions were acquired from Duksan Reagent (Ansan, Republic of Korea). BTO NPs were supplied by US Research Nanomaterials (Houston, TX, USA). Stainless-steel wire meshes were purchased from Mesh Store (Ansan, Republic of Korea). The aluminum tape was obtained from Ducksung (Seoul, Republic of Korea), and the copper tape was supplied by Samwon (Hwasung, Republic of Korea).

2.2 Fabrication of PVDF/BTO@PDA solution

A solvent comprising 6 and 4 mL of acetone and DMF, respectively, was prepared. The process of functionalizing

the BTO NPs using PDA was based on a previous study [40]. The PDA-coated BTO NPs (mass fraction, $\omega = 0$–10 wt%) were ultrasonically dispersed in 8.8 g of the solvent for 30 min. After ultrasonication, 1.2 g of PVDF powder was added and dissolved by stirring at 60 ℃ for 3 h to obtain a PVDF/BTO@PDA solution.

2.3 Electrospinning and Fabrication of TENGs

PVDF/BTO@PDA NFs were fabricated using an electrospinning-spray system (ESR200; NanoNC, Seoul, Republic of Korea). The 2.5 cm × 2.5 cm stainless steel wire meshes were used as the collector. The distance between the spinneret and collector was 12 cm, while a 15 μL/min flow rate was applied throughout the electrospinning process. Electrospinning was performed at 14 kV for 20 min. The NFs were heated in an oven at 80 ℃ for 2 h. The NF mat was detached from the mesh collector and adhered to the aluminum tape. A PVDF/BTO@PDA NF mat was used as the negative triboelectric layer, while aluminum tape was used as the electrode. The glass slide was adhered to aluminum tape to act as a substrate. Copper tape was used as the positive triboelectric layer and electrode, and attached to the pushing tester tip to create a cyclic contact separation motion with a contact area of 1 cm × 1 cm.

2.4 Characterization

The surface morphologies of the patterned NF mats and mesh collectors were observed using field-emission scanning electron microscopy (FE-SEM; JSM-6700F, JEOL, Tokyo, Japan). The three-dimensional (3D) morphologies of the patterned NF mats were characterized using a 3D surface confocal laser scanning microscope (LSM 800 MAT, Carl Zeiss, Oberkochen, Germany). A pushing tester (JLPT-100, Junil Tech, Daegu, Republic of Korea) was used to create the cyclic contact separation mode. The performance of the TENG was investigated using a source measurement unit (2612B, Keithley Instruments, Cleveland, OH, USA). The crystallinities of the PVDF/BTO and PVDF/BTO@PDA NF mats were analyzed using Fourier transform infrared (FTIR) spectroscopy (Nicolet iS50, Thermo Fisher Scientific, Waltham, MA, USA). The surface potential was measured using the Kelvin probe force microscopy (KPFM) [41] mode in an atomic force microscopy (AFM; XE-150, Park Systems, Suwon, Republic of Korea) interfaced with an external lock-in amplifier (RS844, Stanford Research Systems, Sunnyvale, USA). The surface potential was measured while applying a 1 V AC voltage to the probe, with the frequency and phase set to 17 kHz and 0º, respectively, in the lock-in amplifier. A NSC36/Cr-Au probe (radius < 35 nm and 0.6 N/m spring constant) was used for the surface potential measurements. The PDA layers of the BTO@PDA NPs were observed using transmission electron microscopy (TEM; JEM-2010, JEOL). The chemical compositions of the PDA layers were analyzed using X-ray photoelectron Spectroscopy (XPS; K-Alpha+, Thermo Fisher Scientific). The acquired XP spectra were fitted using the Advantage Data System (Thermo Fisher Scientific) and CasaXPS (Casa Software, Teignmouth, UK).

2.5 Signal Processing

Fast Fourier transform (FFT) was performed using MATLAB R2021b software. The fundamental frequency was 0.5 Hz as the cyclic contact separation motion was conducted at 0.5 Hz. The amplitude signals were normalized between 0 and 1. The original length of the frequency axis was 500 Hz (Nyquist frequency) because the sampling rate was set to 1000 Hz.

3 Results and Discussion

3.1 Morphological Characteristics of Patterned Nanofiber Mats

Figure 1 illustrates the electrospinning process for fabricating patterned PVDF/BTO@PDA NF mat using an electroconductive stainless steel wire mesh as the collector. The surface of the BTO NPs was functionalized using a PDA coating to improve their dispersion in the PVDF matrix. Further, the PVDF/BTO@PDA solution was electrospun in a mesh collector. During electrospinning, a high voltage applied to the spinneret created an electric field that electrified the solution. The charged solution was ejected from the needle tip and transformed into polymeric NFs during the whipping process along the electric field [26, 42]. The deposition process of the NFs was influenced by the electric field and Coulomb interactions between the positive charges on the NFs and negative charges on the collector induced by the NFs [43, 44]. As the NFs approached the collector, strong Coulomb interactions occurred between the NFs and protruding wires owing to the reduced distance than the other low-lying areas of the mesh collector [35, 37]. The protruding wires exhibited a high density of deposited NFs, whereas the voids exhibited a lower NF density. After electrospinning the NFs for 20 min, the mats were detached from the mesh collector. The surface of the NF mats in contact with the mesh collector formed a uniquely patterned morphology.

The FE-SEM images of the mesh collector are shown in Fig. 2a–c. In this study, three distinct meshes (100, 300, and 500) were used. The mesh number is a quantitative indicator, which represents 10 000, 90 000, and 250 000 openings per square inch for the 100, 300, and 500 meshes, respectively. With an increase in the mesh number, the fineness

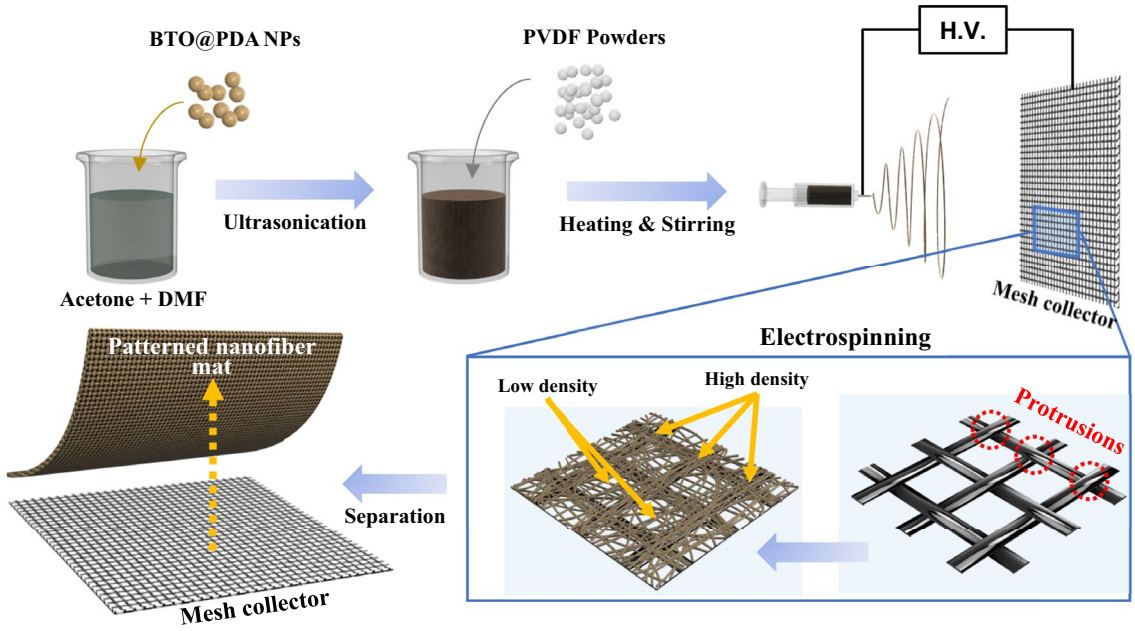


Fig. 1 Schematic of the fabrication process of a patterned PVDF/BTO@PDA nanofiber mat using an electroconductive mesh collector

of the mesh collector increased, whereas the diameter of the wires decreased. The interwoven structure of the wires determined the morphology of the mesh collectors. The 100 mesh comprised a plain weave, whereas the 300 and 500 meshes comprised twill weaves, which are generally used for finer meshes owing to manufacturing preferences [35]. In this study, there is no particular reason for using different weaves according to the mesh size. Protruded wires ("Protrusions"; yellow boxes in Fig. 2a) appeared repeatedly in the horizontal and vertical directions owing to the interweaving of the wires. Further details on the protrusions for each type of mesh collector are presented in Table S1 (Supplementary Information).

Figure 2d–f and g–i show the FE-SEM images of the PVDF/BTO@PDA NF mats with 5 wt% BTO@PDA NPs after being detached from the 100-, 300-, and 500-mesh collectors at 100 × and 300 × magnifications, respectively. The regions of the NF mat that were in contact with the protrusions created distinct patterns. The NFs formed on a single protrusion were referred to as a "pattern". NFs tended to accumulate more densely onto protrusions owing to their strong Coulombic forces, thereby resulting in varying NF densities between the protrusions and voids on the mesh collector and forming a patterned NF mat. The patterns as portrayed in Fig. 2d (marked with yellow boxes) demonstrate that the configuration of the NF patterns corresponded with that of the protrusions. Additionally, the fineness of the metal wires increased with the mesh number. Hence, the morphology patterns formed using the 500-mesh collector exhibited the highest density.

The confocal laser scanning microscopy images of the patterned NF mats using 100-, 300-, and 500-mesh collectors are shown in Fig. 2j–l, whereas their corresponding 3D images are shown in Fig. S1 (Supplementary Information). The NF mat exhibited patterns with varying heights, thereby forming 3D structures. The patterns formed with the 100-mesh collector had the largest lateral size and more wire protrusions owing to the larger wire diameter than the other mesh collectors. Consequently, the pattern heights of 20–28, 12–18, and 9–13 μm were obtained using the 100-, 300-, and 500-mesh collectors, respectively. Figure 2d–l shows the surfaces of the NF mats that were in direct contact with the mesh collector after detachment. The NF morphology was preserved regardless of the electrospinning time.

3.2 Enhanced Performance Using Nonadditive Physical Modification

The distinct morphologies of the patterned NF mats induced by the three types of mesh collectors could result in different triboelectric effects when used as contact layers in TENGs. To investigate the morphological influence of the patterned NF mats on the TENG performance, pure PVDF NFs were used as negative triboelectric materials. The structure and operational principle of the TENG are illustrated in Fig. 3a. Copper tape was selected as the positive triboelectric material owing to its electron-losing ability [45]. In the adopted TENG structure, the charge-generation mechanism operates on a periodic contact–separation mode. When two triboelectric materials come into contact

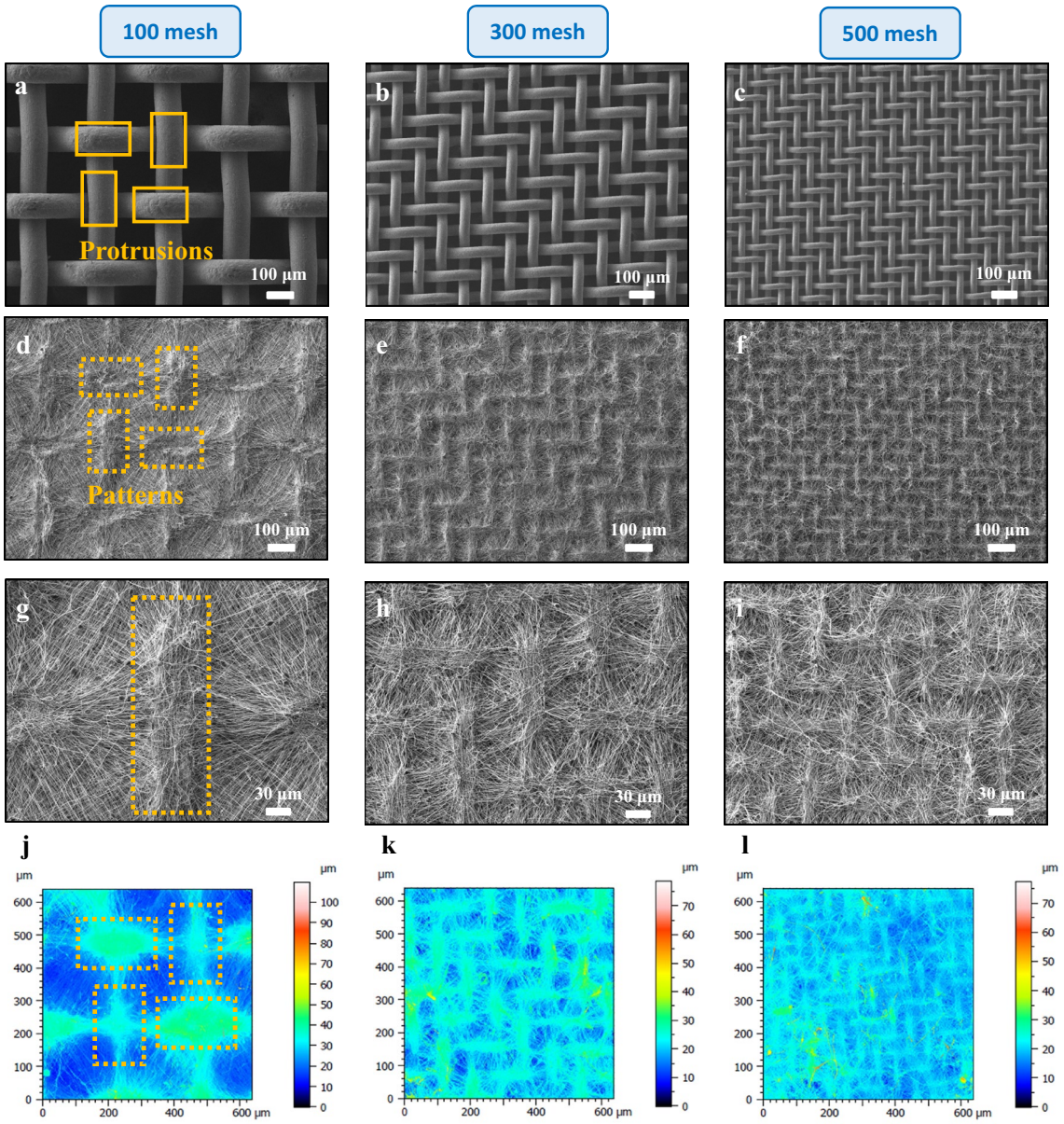


Fig. 2 FE-SEM images of the stainless steel wire mesh collectors: **a** 100, **b** 300, and **c** 500 meshes. FE-SEM images of the PVDF/BTO@PDA NF mat from **d** 100-, **e** 300-, and **f** 500-mesh collectors at a 100×magnification. FE-SEM images of the mat from **g** 100-, **h** 300-, and **i** 500-mesh collectors at a 300×magnification. Confocal laser scanning microscope images of the patterned PVDF nanofiber mat from the **j** 100-, **k** 300-, and **l** 500-mesh collectors

under an external mechanical force, electrons transfer from the copper tape to the patterned PVDF NF mat owing to the differences in the electron affinities ((i) contact). Hence, the copper contact layer was positively charged, whereas the PVDF NF mat was negatively charged. When the copper contact layer was separated from the patterned PVDF NF mat, electrons flowed from the bottom electrode to the top electrode ((ii) contact-to-separation). When the triboelectric pair reached the maximum separation distance, the charge transfer between the electrodes ended because the charge induction was completed ((iii) separation). The electrons flowed inversely when the two contact layers approached each other again ((iv) separation-to-contact). Consequently, a cyclic alternating voltage signal was acquired from the periodic contact–separation mode, as shown in Fig. 3b. As the NF mat exhibited piezoelectric properties, a specific amount of electrons flowed through the external circuit owing to the piezoelectric potential. The working mechanism of the PENG is portrayed in Fig. S2.

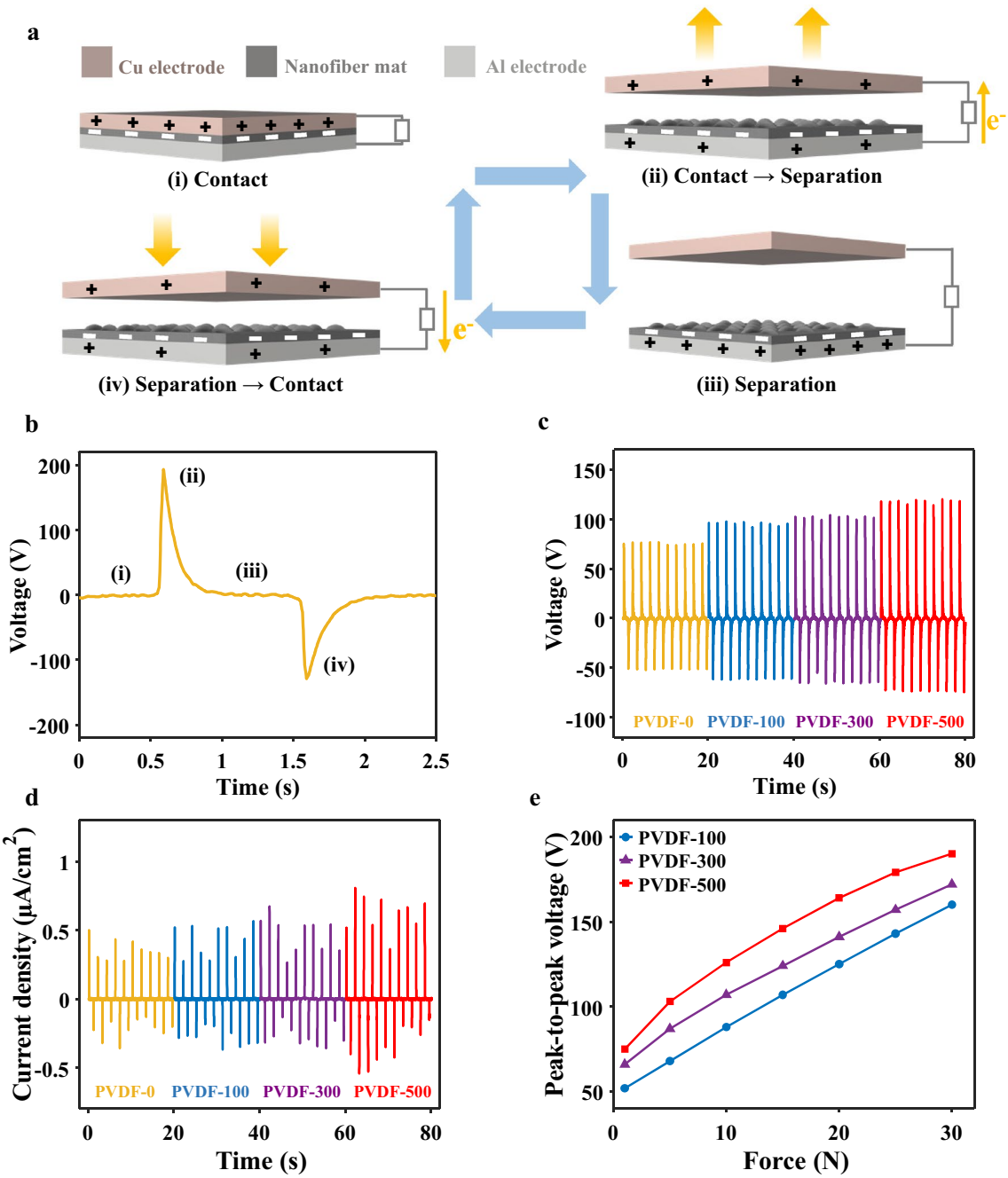


Fig. 3 **a** Structure and working principle of the TENG, **b** voltage signal of the TENG, **c** voltage signals of the PVDF-based TENG obtained using collectors of different mesh sizes, **d** current densities of the PVDF-based TENG obtained using collectors of different mesh sizes, **e** peak-to-peak voltage of the three mesh collectors relative to the applied force

In this study, different types of NF mats were produced, and their details are presented in Table 1. Figure 3c and d show the variations in the output voltages and current densities of the TENGs, respectively, with PVDF-0, PVDF-100, PVDF-300, and PVDF-500 contact layers under a force and frequency of 30 N and 0.5 Hz, respectively. An aluminum plate, instead of a mesh collector, was used for the nonpatterned NF mat (PVDF-0). The morphology of the NF mat considerably impacted the TENG performance. The patterned NF mats yielded superior TENG performance than the nonpatterned NF mat. Moreover, the voltage and current signals increased as the number of meshes increased. The average peak-to-peak voltage (V_{pp}) signals of the TENG with the PVDF-0, PVDF-100, PVDF-300, and PVDF-500 contact layers were 126.7, 159.6, 172.3, and 190.9 V, respectively. To comprehend the relationship between the heightened

Table 1 Notations of the NF mats based on the type of the collector and solution

Solution Type of collector	PVDF	PVDF/BTO	PVDF/BTO@PDA
Aluminum plate	PVDF-0	PVDF/BTO-0	PVDF/BTO@PDA-0
100 mesh	PVDF-100	PVDF/BTO-100	PVDF/BTO@PDA-100
300 mesh	PVDF-300	PVDF/BTO-300	PVDF/BTO@PDA-300
500 mesh	PVDF-500	PVDF/BTO-500	PVDF/BTO@PDA-500

voltage signals and morphology of the NF mats, the approximate total surface area of the patterns in a 1 in × 1 in area was estimated. The morphology of a single pattern resembled a half-prolate spheroid as shown in Fig. 2j–2l. The surface area of a prolate spheroid can be calculated if its equatorial and polar radii are known [46]. The approximate surface area of the NF pattern was computed assuming a prolate spheroid pattern shape. The detailed computational process is shown in Fig. S3 and Table S2. The total surface area of the patterns for the same size of mesh collectors (1 in × 1 in) increased with the number of meshes. The largest surface area of the NFs deposited onto the protrusions was recorded in the 500-mesh collector, followed by the 300- and 100-mesh collector. Generally, a contact layer with a large surface area is preferred for TENGs because it enhances the effective contact area during contact and separation motions, thereby resulting in the generation of a larger surface charge [47].

Figure 3e shows the force sensitivity test results in response to an applied force of 30 N for three NF mats. The highest V_{pp} was generated from the NF mats fabricated using a PVDF-500, whereas the lowest V_{pp} was observed using PVDF-100 over the entire force range. Additionally, the V_{pp} of the NF mats increased with an increase in the applied force. When a small external force was applied to the patterned NF mat, only the upper section of the pattern was in contact, thereby resulting in a lower output voltage. However, a larger force caused a considerable deformation of the patterns, thereby increasing the surface charge (Fig. S4) [48]. Furthermore, none of the NF mats exhibited saturation in an output voltage under the applied force of 30 N, thereby indicating that the patterned NF mats did not fully deform even under substantial force [49]. However, the rate of V_{pp} increase varied among the PVDF-100, PVDF-300, and PVDF-500 mats. The V_{pp} of PVDF-100 increased linearly with an increase in the force, whereas that of PVDF-300 and PVDF-500 exhibited a nonlinear relationship with the force. Notably, the rate of V_{pp} increase in PVDF-500 gradually decreased with an increase in the force. The different rates of V_{pp} increase were attributed to the different pattern sizes. Smaller pattern sizes resulted in larger output voltages induced by smaller external forces, thereby benefitting from a larger pattern density. However, as the applied force increased, the output voltage did not increase owing to the small size of the individual patterns. Fig. S5 shows the output voltage of the TENG relative to the changes in the total surface area of patterns (Table S2) for a force range of 10–30 N. As the total surface area of patterns increased, the output voltage of the TENG increased, irrespective of the applied force.

3.3 Enhanced Performance Using Additive Physical Modification

Additive physical modification is an effective approach for enhancing the performance of TENG in conjunction with nonadditive physical modifications. In this study, a PVDF/BTO NPs solution was electrospun using 500-mesh collectors and 400 nm BTO NPs to induce a synergetic effect between the two physical modification methods. The effect of the PDA-functionalized BTO NPs on the TENG performance was investigated by separately incorporating bare BTO NPs and BTO@PDA NPs into the PVDF. The TEM images shown in Fig. S6a and S6b demonstrate the successful coating of a PDA layer with a thickness of approximately 6.9 nm on the surface of the BTO NPs. The presence of the PDA layer was also confirmed using XPS (Fig. S6c).

The TENG performance of PVDF/BTO-500 and PVDF/BTO@PDA-500 with different filler concentrations (0, 1, 3, 5, 7, and 10 wt%) is shown in Fig. 4a and b, respectively. The V_{pp} of PVDF/BTO-500 increased from 190 to 274 V as the BTO NP concentration increased from 0 to 5 wt%. However, the voltage decreased to 208 V after the incorporation of 10 wt% BTO NPs. The output voltage of PVDF/BTO@PDA-500 exhibited a trend similar to that of PVDF/BTO-500. However, its TENG performance was notably improved. As the BTO@PDA NP concentration increased to 5 wt%, the voltage signals of PVDF/BTO@PDA-500 increased to 318 V. However, the voltage signals decreased to 223 V after adding 10 wt% BTO@PDA NPs. The current density of the TENGs based on PVDF/BTO-500 and PVDF/BTO@PDA-500 is shown in Fig. S7a–b. Owing to the uneven current signals, the transferred charge was calculated based on the current signal. As shown in Fig. S7c–d, the maximum transferred charge was observed at 5 wt%, regardless of the presence of PDA coating on the BTO NPs. The transferred charge increased when the BTO NPs were coated with PDA.

The crystallinity of PVDF is crucial to PENGs and TENGs, thereby simultaneously enhancing the piezoelectric properties and charge transfer between PVDF and copper [29, 50]. FTIR spectroscopy was used to investigate

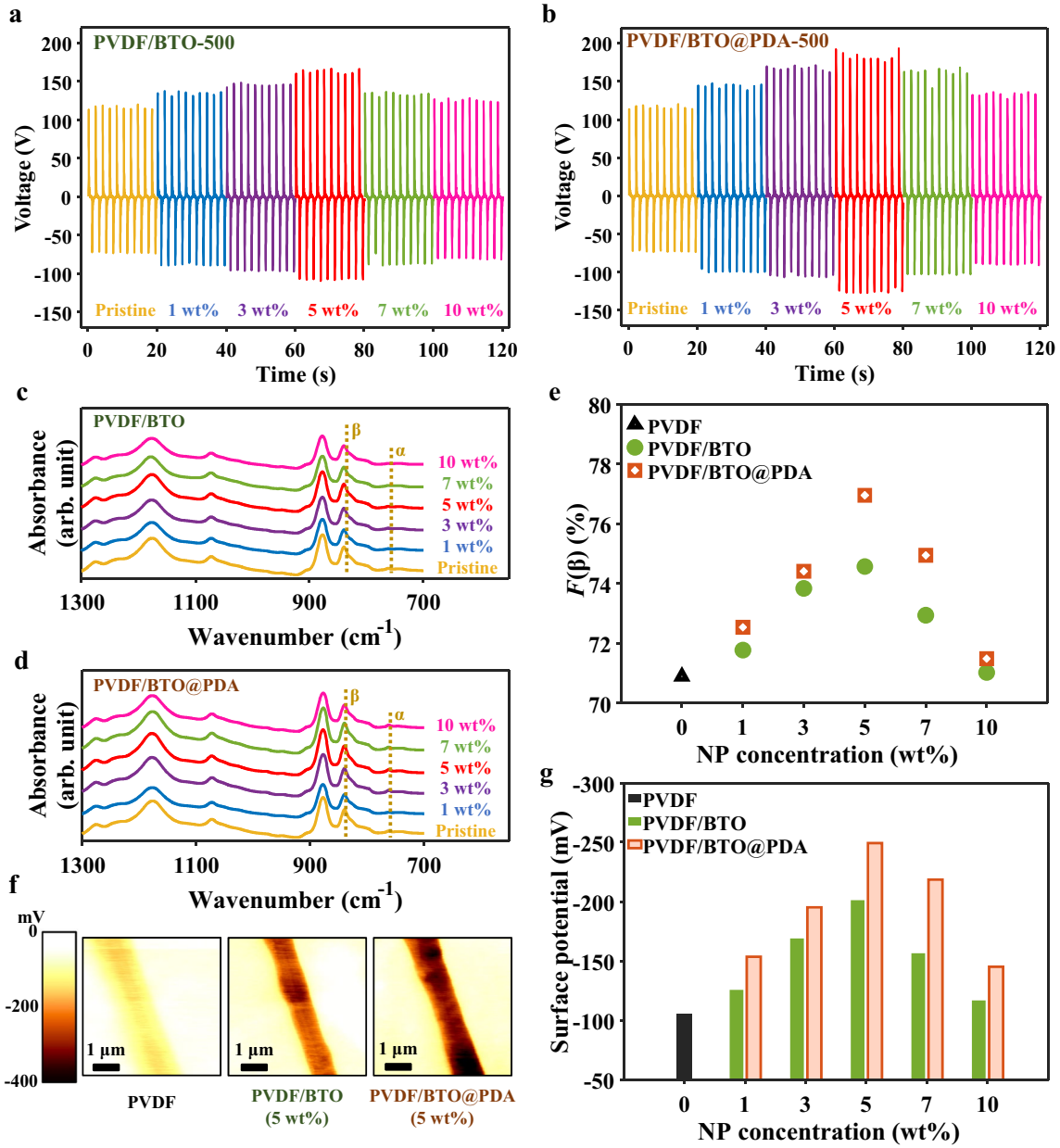


Fig. 4 Output voltage of the TENG with **a** PVDF/BTO-500 NFs and **b** PVDF/BTO@PDA-500 NFs relative to the NP concentration. FTIR spectra of **c** PVDF/BTO NFs and **d** PVDF/BTO@PDA NFs at NP concentrations. The wavelengths at 762 and 840 cm^{-1}, which correspond with the α- and β-phases, respectively, are represented by the dotted lines. **e** β-phase fraction of PVDF/BTO and PVDF/BTO@PDA NFs relative to the NP concentration. **f** Surface potential images of PVDF, PVDF/BTO, and PVDF/BTO@PDA NFs measured using KPFM. **g** Surface potential of PVDF/BTO and PVDF/BTO@PDA NFs relative to the NP concentration

the crystallinity changes in PVDF upon the addition of BTO and BTO@PDA NPs. Figure 4c and d shows the FTIR spectra of the PVDF/BTO and PVDF/BTO@PDA NF mats, respectively, at different filler concentrations (0, 1, 3, 5, 7, and 10 wt%). The β-phase fraction ($F(\beta)$) was expressed using Eq. (1), assuming that the IR absorption adhered to the Lambert–Beer law as follows [51]:

$$F(\beta) = \frac{A_\beta}{\left(K_\beta / K_\alpha\right)A_\alpha + A_\beta} \quad (1)$$

where A_α and A_β denote the absorbance of the vibration peaks at 762 and 840 cm^{-1}, respectively [52], and K_α (6.1×10^4 cm^2/mol) and K_β (7.7×10^4 cm^2/mol) the absorption coefficients at 762 and 840 cm^{-1}, respectively. $F(\beta)$ was

computed for the NF mats containing 0–10 wt% BTO and BTO@PDA NPs, as shown in Fig. 4e. The $F(\beta)$ of PVDF/BTO NF mat gradually increased from 70.8 to 74.5% as the BTO NP concentration increased from 0 to 5 wt%. The crystallinity of the PVDF was improved by incorporating BTO NPs, which served as nucleation sites for the β-phase [39]. The resulting polar phases in the crystallization region improved the charge transfer between the NF mat and copper, thereby enhancing the TENG performance [29]. However, $F(\beta)$ of PVDF/BTO decreased to 71.0% after adding 10 wt% BTO NPs. This could be attributed to the aggregation of BTO NPs, which hampered the transformation of the PVDF chains into an all-trans configuration when a high content of BTO NPs was incorporated [53]. As the content of the BTO NPs increased, the density of the interface hole defects between the BTO NPs and the PVDF matrix increased, thereby impeding the increase in $F(\beta)$ of the PVDF/BTO matrix [54]. Conversely, PVDF/BTO@PDA exhibited a superior $F(\beta)$ than PVDF/BTO across the range of NP concentrations. The highest $F(\beta)$ (76.9%) was observed for the PVDF/BTO@PDA NF mat containing 5 wt% BTO@PDA NPs. Thus, the crystallinity of the PVDF was further enhanced by functionalizing the surface of the BTO NPs with PDA. During electrospinning, intermolecular interactions between the positively charged amine groups of PDA and negatively charged difluoromethylene groups of PVDF formed strong hydrogen bonds, thereby enhancing the crystallinity of the PVDF chains and β-phase content [55, 56]. Additionally, the improved dispersion of the BTO NPs in the PVDF matrix using PDA coating enhanced the formation of β-phase in the PVDF [57, 58]. However, when 7 and 10 wt% NPs were incorporated, $F(\beta)$ decreased to 74.9% and 71.4%, respectively. This could be attributed to the increase in the number of residual interface hole defects between BTO@PDA and the PVDF matrix [59], which disrupted the formation of the symmetrical β-phase configuration [54].

The β-phase of the NF decreased when the BTO content exceeded 5 wt%. Previous studies on PVDF/BTO composite films showed that the output voltage of the TENG or β-phase fraction increased at higher BTO contents than 5 wt% [38, 60]. However, previous studies on electrospun PVDF/BTO NFs showed that the maximum output voltage from the TENG was observed at a BTO content of 5 wt%, which corresponds with the findings of this study [61]. These differences in the optimal BTO contents can be attributed to the electrospinning process. As the BTO content increased in the PVDF/BTO solution, the NFs became more brittle owing to the increase in the concentration of ceramic particles. This increase in the brittleness can result in fiber damage and fracture owing to the mechanical stretching and whipping force during electrospinning, thereby hindering the formation of β-phase in PVDF. Further, electrospinning can trap the BTO NPs within the NFs as shown in Fig. 5, thereby hindering their uniform distribution than film structures. This trapping effect can further promote the aggregation of BTO NPs, thereby reducing the crystallinity of PVDF.

The addition of ceramic fillers to the PVDF matrix affected the surface potential of the contact layer. A considerable surface potential difference between the two contact layers enhanced the surface charge density of the TENG via contact electrification [62, 63]. Hence, KPFM was used to study the effect of BTO and BTO@PDA NP concentrations on the surface potential of the contact layers. Figure 4f shows the surface potential images of pristine PVDF, PVDF/BTO, and PVDF/BTO@PDA NFs with 5 wt% NPs. Fig. S8 portrays the surface potential images of NFs with 0–10 wt% NPs. The inclusion of the BTO NPs reduced the surface potential, which was further decreased by functionalizing the BTO NP surfaces using PDA. Figure 4g illustrates the surface potentials of PVDF/BTO and PVDF/BTO@PDA at different NP concentrations. The surface potential decreased from − 105.6 to − 201.2 mV after incorporating 5 wt% BTO NPs. After modifying the surface of the BTO NPs with the PDA layer, the surface potential further decreased to − 249.4 mV. However, it increased for both PVDF/BTO and PVDF/BTO@PDA when the NP concentration exceeded 5 wt% owing to the NP agglomeration. The surface potential measurement results reveal a similar trend to the total transferred charge previously calculated in Fig. S7e. This shows the close relationship between the surface potential and charge density of the TENGs.

The impact of the BTO NP size on the performance of TENG was analyzed. The output voltages of PVDF/BTO-500 and PVDF/BTO@PDA-500 with various particle sizes (100, 200, 300, and 400 nm) and 5 wt% BTO NPs are shown in Fig. 5a and b, respectively. As the size of the BTO NPs increased from 100 to 400 nm, the V_{pp} of PVDF/BTO-500 increased from 234 to 274 V. Similar trends were observed in the output voltage of PVDF/BTO@PDA-500 and PVDF/BTO-500 wherein the output voltage increased with an increase in the NP size. After coating the BTO NPs with the PDA layer, the V_{pp} increased further, thereby reaching 283 and 315 V for NP sizes of 100 and 400 nm, respectively. The extend of the voltage increase owing to the PDA coating varied with the NP size. As the NP size decreased, the rate of voltage increase rose. Notably, the output voltage increased by 20.9, 18.9, 15.0, and 14.9% with the surface modification of 100-, 200-, 300-, and 400-nm BTO NPs, respectively, with PDA.

Figure 5c and d show the FE-SEM images of the PVDF/BTO and PVDF/BTO@PDA NFs with different NP sizes, respectively. The agglomeration of BTO NPs was observed in the PVDF/BTO NFs, regardless of the BTO NP size. Conversely, the NP dispersion considerably improved in the PVDF/BTO@PDA NFs owing to the enhanced interfacial compatibility between PVDF and the NPs. Hence,

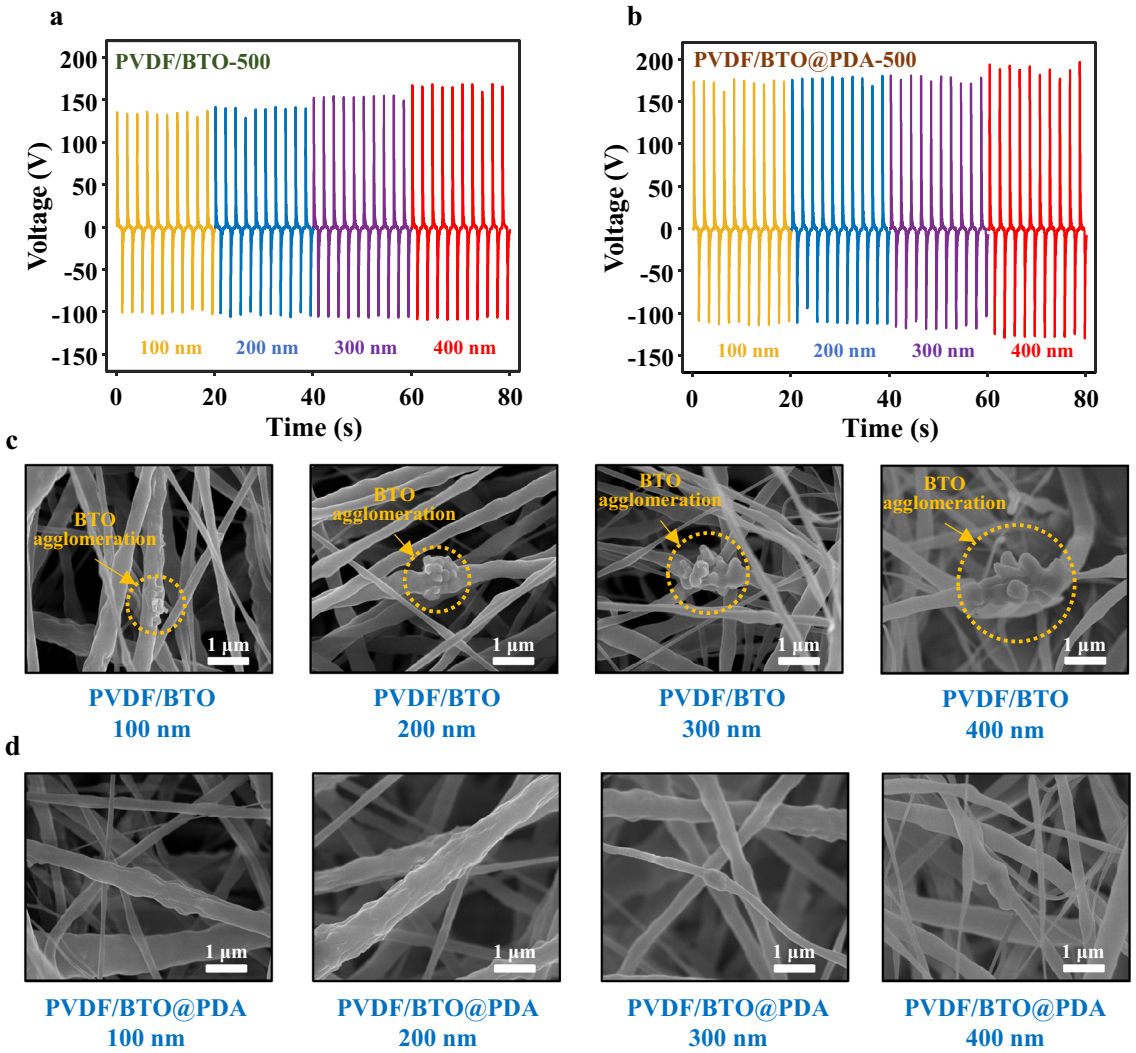


Fig. 5 Voltage signals of the TENG based on the **a** PVDF/BTO-500 and **b** PVDF/BTO@PDA-500 NFs relative to the BTO NP size. FE-SEM images of **c** PVDF/BTO NFs and **d** PVDF/BTO@PDA NFs of different NP sizes

the considerable improvement in the TENG performance achieved by functionalizing the surface of the BTO NPs was mainly attributed to the improved degree of the NP dispersion. When a similar NP concentration was incorporated, the number of NPs decreased as the NP size increased. Consequently, the effective interfacial area between the BTO NPs and PVDF was increased by functionalizing the surface of the BTO NPs, particularly for small NPs. When the NPs were functionalized by PDA, the largest increase in the output voltage of the 100-nm NPs can be explained by the largest effective interfacial area between the NPs and the PVDF. As shown in the FE-SEM images (Fig. 5c and 5d), the diameter of the NFs ranged from 100 nm to 1 μm. Small NPs tended to be encapsulated within the NFs. Conversely, large NPs tended to protrude from the NF, thereby increasing the surface roughness of the NFs and producing effects that are similar to those of nanopatterning. Hence, the highest voltage signals of the TENG with the PVDF/BTO@PDA-500 contact layer with 400-nm NPs can be attributed to the increase in the effective contact area.

3.4 Synergetic Effects of Nonadditive and Additive Physical Modifications

To date, the use of a mesh collector for electrospun PVDF NFs remains a facile method to induce the synergetic effects of additive and nonadditive physical modifications on the contact layer of TENG. The progressive increases in the output voltage and current density of the TENG via the adoption of physical modifications are shown in Fig. 6a and b, respectively. Figure 6c shows a schematic of the progressive development of the contact layer structure. First, the TENG fabricated by electrospinning PVDF on a flat collector generated the smallest V_{pp} (126 V) and peak-to-peak current

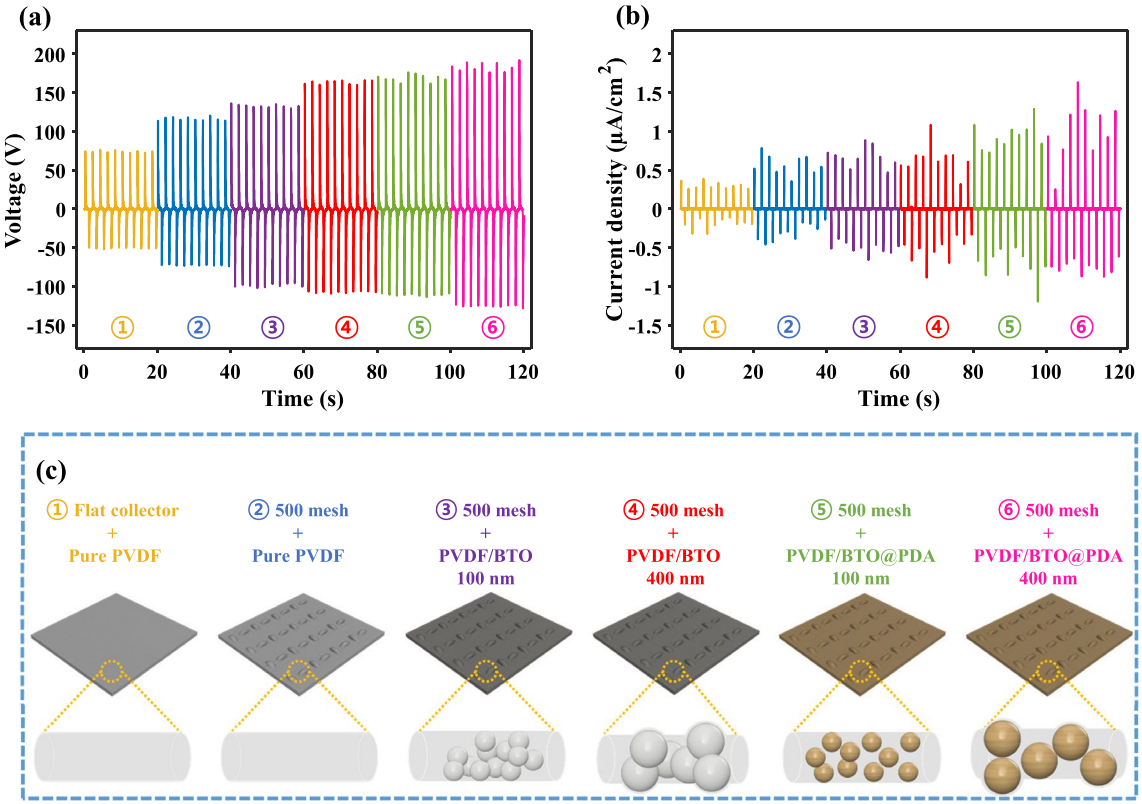


Fig. 6 **a** Output voltage signals and **b** current density of the TENG with six different friction layers. **c** Schematic of the PVDF NF mats physically modified using additive and nonadditive methods: ① pure PVDF NF mat from a flat collector, ② pure PVDF NF mat from the 500-mesh collector, ③ PVDF/BTO (100-nm NPs) NF mat from the 500-mesh collector, ④ PVDF/BTO (400-nm NPs) NF mat from the 500-mesh collector, ⑤ PVDF/BTO@PDA (100-nm NPs) NF mat from the 500-mesh collector, and ⑥ PVDF/BTO@PDA (400-nm NPs) NF mat from the 500-mesh collector

density J_{pp} (0.56 μA/cm^2) (No. 1). With the use of a 500-mesh collector, micropatterns were introduced on the PVDF NF mat, thereby increasing V_{pp} and J_{pp} to 190 V and 0.96 μA/cm^2, respectively (No. 2). The performance of TENG was further enhanced by incorporating BTO NPs into the PVDF matrix owing to the large piezoelectric coefficient of the BTO NPs and improved crystallinity of the PVDF. V_{pp} (234 V) and J_{pp} (1.31 μA/cm^2) increased with the use of 100-nm BTO NPs (No. 3). Utilizing 400-nm BTO NPs further increased V_{pp} (274 V) and J_{pp} (1.44 μA/cm^2) by nanopatterning the surface of the PVDF NFs (No. 4). The TENG performance was further enhanced by functionalizing the surface of the BTO NPs with PDA, thereby enabling the uniform dispersion of the BTO NPs and increasing the crystallinity of the PVDF. The V_{pp} and J_{pp} of PVDF/BTO@PDA with 100-nm BTO NPs were 283 V and 1.88 μA/cm^2, respectively (No. 5), while those with 400-nm BTO NPs were 315 V and 2.05 μA/cm^2, respectively (No. 6). The synergetic effect of the physical modifications increased the voltage by 249% from 126 to 315 V. Notably, the performance of the ferroelectric-assisted TENG fabricated in this study was better than the previously reported PVDF-based TENGs given that the performance was evaluated at a force of 30 N and a frequency of 0.5 Hz (Table S3).

3.5 Identification of Triboelectric Signals Using Fast Fourier Transform

An NF mat comprising PVDF and piezoelectric ceramic fillers was used as the contact layer for the TENGs. During the contact–separation motion, the piezoelectric and triboelectric effects occurred simultaneously in the TENGs. As such, the contribution of the piezoelectric effect in the TENGs cannot be easily identified because the output signal was combined with the signal originating from the triboelectric effect. Moreover, the amplitudes of the voltage and current signals owing to the piezoelectric effect are smaller than those caused by the triboelectric effect. Hence, the cyclic compression under preloading conditions considerably reduced the influence of triboelectricity on the output signals. However, the triboelectric effect of a porous contact layer should not be completely ignored when using this

experimental method because the external force could alter its internal geometry, thereby resulting in an inconsistent contact between the contact layer and electrode [64].

FFT was employed to distinguish the piezoelectric and triboelectric effects on the output voltage signals under various experimental conditions [65]. FFT is a computational method used to determine the discrete Fourier transform of a sequence. It has been extensively utilized in diverse engineering fields [66–68], and has been recently applied in energy harvesting devices to decouple output signals originating from the TENGs and PENGs [65]. As illustrated in Fig. 7a, three contact layers were prepared: polydimethylsiloxane (PDMS), poled PVDF, and patterned PVDF-500 mats. The PVDF film was electrically poled at 30 MV/m for 1 h. The pushing test was conducted by applying a preload force of 10 N, accompanied by a cyclic compression force of 30 N, to each contact layer at a frequency of 0.5 Hz. The obtained output voltages are shown in Fig. 7b and their respective frequency domain signals acquired using FFT are shown in Fig. 7c. As shown in Fig. 7b(i), the output voltage signals of the PDMS film were almost similar to the noise level because PDMS is not a piezoelectric material. In the corresponding frequency domain signal as shown in Fig. 7c(i), only the DC offset component was observed, thereby indicating that there was no periodic variation in the voltage signals. Hence, the frequency domain signal patterns for the nonpiezoelectric materials subjected to a pushing test, excluding the triboelectric effect, were confirmed. Contrarily, the periodic voltage signals of $V_{pp} = 0.04$ V were observed from the poled PVDF film under similar experimental conditions, as shown in Fig. 7b(ii). The periodic variations in the voltage signal were clearly captured by the frequency component of < 10 Hz in the frequency domain as shown in Fig. 7c(ii). The observed frequency components originated from the piezoelectric effect because no triboelectric effect was observed in the PDMS film under similar conditions. Finally, the patterned PVDF NF mat yielded the highest output voltage ($V_{pp} = 0.65$ V), as shown in Fig. 7b(iii). The corresponding frequency domain signal displayed a frequency component of < 33 Hz, as shown in Fig. 7c(iii). Unlike the PVDF film, the appearance of high-frequency components (10–33 Hz) in the patterned PVDF NF mat can be attributed solely to the triboelectric effect. As shown in Fig. 7a(ii) and (iii), the pushing tests were conducted at the same pushing frequency of 0.5 Hz. If only the piezoelectric effect occurred in (iii), then Fig. 7c(ii) and (iii) would exhibit similar forms. Hence, additional deformation occurred at the protruding regions of the NF mat during cyclic compression, even under preloading conditions, thereby generating a triboelectric effect. When conducting a pushing test with a force of 30 N without preload, the triboelectric effect considerably contributes to the output voltage signals once the initial contact is made between the pushing tip and protruding regions in the NF mat. As shown in Fig. S9, the voltage signals can be increased by reducing the preloading conditions (1 N). In the contact separation motion, reducing the preloading force enhances the piezoelectric effect while simultaneously improving the triboelectric effect between the patterned NF mat and Cu electrode. Hence, distinguishing between the contributions of the piezoelectric and triboelectric effect to the output voltage signals was not very feasible. Notably, the enhanced β-phase content of the PVDF NFs during electrospinning was also partially responsible for the considerable increase in the voltage signal as shown in Fig. 7b(iii).

The PVDF NF mat generated an output voltage through the triboelectric effect, even under preload conditions, owing to the friction within its internal geometry. Subsequently, the triboelectricity arising between the PVDF NF mat and positive triboelectric material (copper) during the contact–separation motion was clarified. As shown in Fig. 7d, the output voltage was measured by varying the separation distance (1, 3, 5, and 7 mm) during the contact–separation motion. V_{pp} increased from 93 to 126 V as the distance increased from 1 to 7 mm (Fig. 7e). A larger separation distance between the two electrodes increased the potential difference, thereby increasing the output voltage or current signal [69–71]. The enhanced triboelectric effect induced by a larger separation distance can be confirmed in the frequency domain by the increase in the spectral bandwidth [65]. Hence, the time-domain signals as shown in Fig. 7e were converted into frequency domain signals using FFT (Fig. 7f). The termination point ($f_{3\%}$) was set at a frequency corresponding to 3% of the normalized intensity at the fundamental frequency ($f_0 = 0.5$ Hz). The spectral bandwidth was expressed as the frequency span between the termination point and fundamental frequency ($f_{3\%}$–f_0). The spectral bandwidth increased from 17.5 to 22.5 Hz as the separation distance increased from 1 to 7 mm, thereby confirming that the voltage signal originated from the triboelectric effect between the two contact layers. Hence, the general characteristics of TENG are also observed in ferroelectric-assisted TENG based on a patterned NF mat.

3.6 Application of PVDF/BTO@PDA Nanofiber Mats in Energy Harvesting Devices

Numerous tests were conducted to assess the practical applicability of the TENGs. The maximum power density of the TENG with PVDF/BTO@PDA-500 (400-nm BTO NPs) was measured by varying an external load (100 kΩ to1 GΩ). The power density (P) was calculated as $P = V_L^2/(R_L \times A)$, where V_L represents the peak-to-peak load voltage, R_L the load resistance, and A the area of the contact layer. As shown in Fig. 8a, V_L increased with R_L, although its increase rate decreased after $R_L = 100$ MΩ. The maximum P

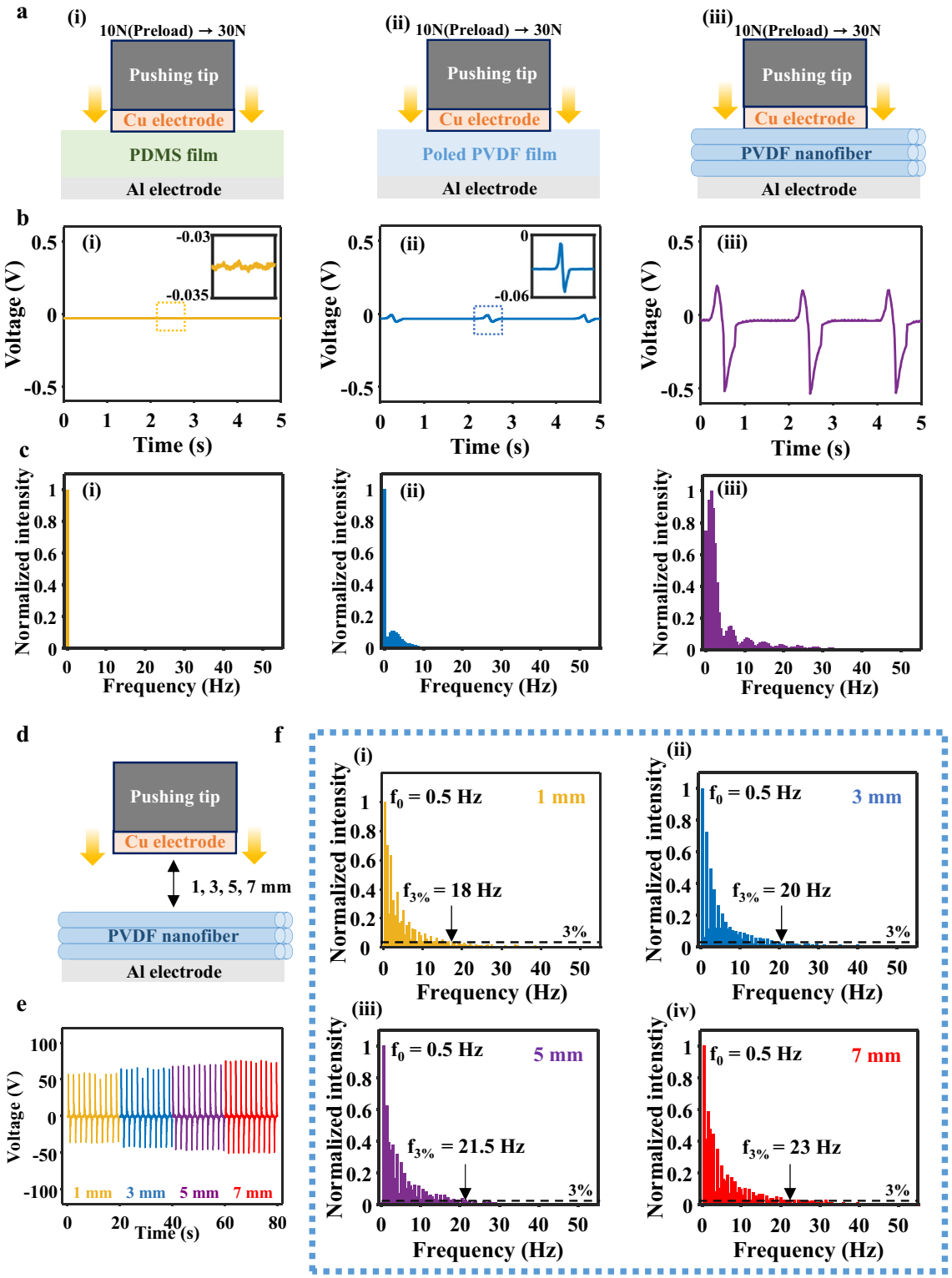


Fig. 7 **a** Illustration of the preloading test with different friction layers: i) PDMS film, ii) poled PVDF film, and iii) PVDF NF. **b** Output voltage of the friction layers under an applied preload. **c** Normalized FFT spectra of the friction layers. **d** Schematic of the contact–separation motion with different separation displacements. **e** Output voltage of the PVDF NF mat with different separation displacements. **f** Normalized FFT spectrum of the PVDF NF mat with different separation distances: i) 1 mm, ii) 3 mm, iii) 5 mm, and iv) 7 mm

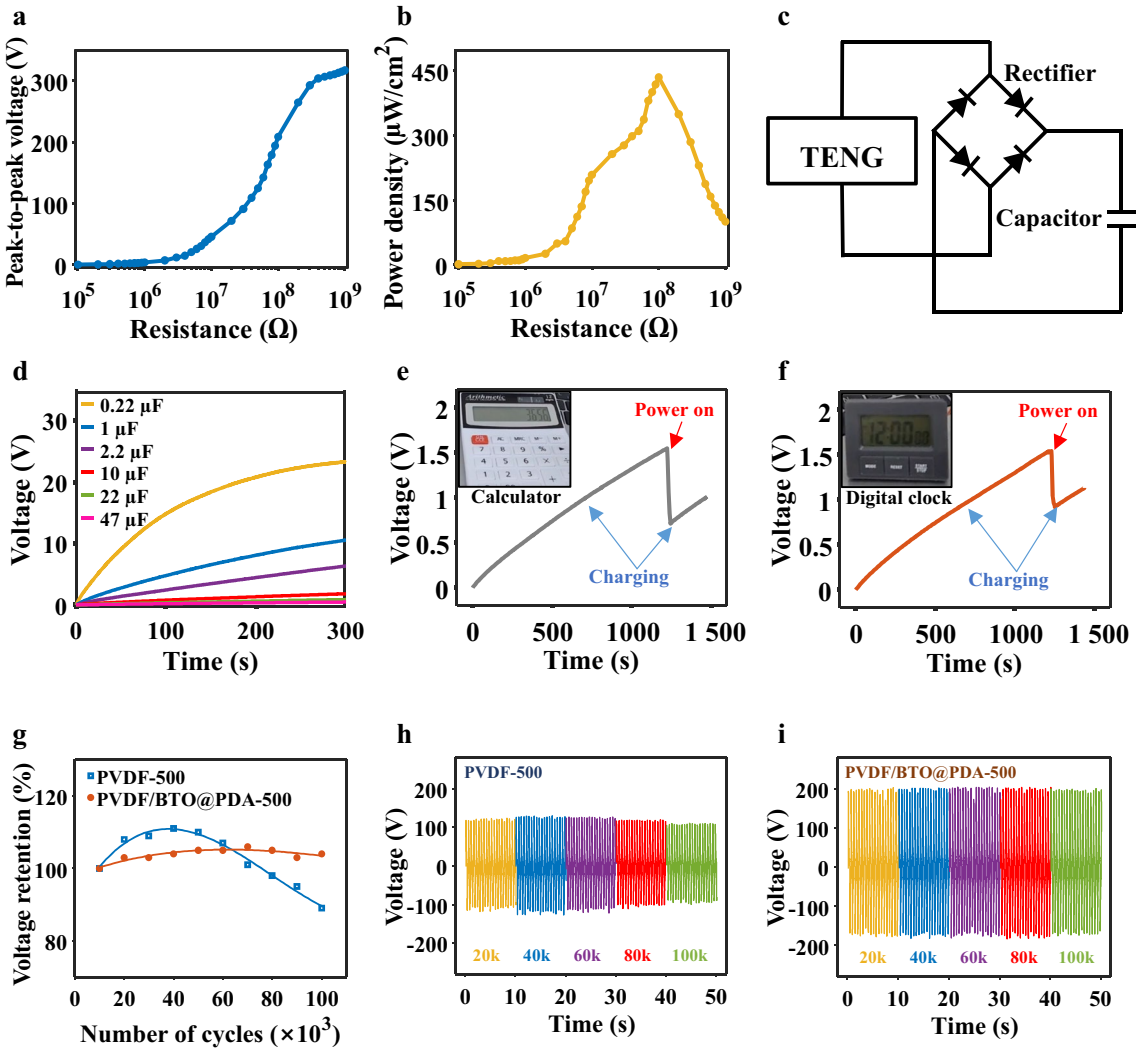


Fig. 8 **a** Output voltage, and **b** power density under varying external resistance. **c** Equivalent circuit diagram of the TENG connected to the external circuit through a rectifier. **d** Charging capacitors with different capacitances using TENGs. **e** Charging and discharging curve of the capacitor to power a calculator, and **f** a digital clock. **g** Mechanical stability test of the TENGs based on the PVDF-500 and PVDF/BTO@PDA-500 NF mats for the 100 000 cycles of contact and separation motions. **h** Output voltage of the TENGs with PVDF-500 and **i** PVDF/BTO@PDA-500 after different contact separation cycles (20 000, 40 000, 60 000, 80 000, and 100 000)

of 434.76 μW/cm^2 was obtained at $R_L = 100$ MΩ (Fig. 8b). To investigate the charging capability of the TENGs, commercial capacitors with capacitances of 0.22, 1, 2.2, 10, 22, and 47 μF were charged using a bridge rectifier. Figure 8c shows a similar circuit diagram of the TENG connected to an external circuit to charge the capacitors. As shown in Fig. 8d, different capacitors were charged at different rates based on the capacitance. Additionally, the feasibility of the TENG was demonstrated by powering the commercial electronics. The calculator (Fig. 8e and Video S1) and digital clock (Fig. 8f and Video S2) were successfully powered by charging the 47 μF capacitor.

The reliability and the stability of TENG devices are crucial for practical applications. To assess the mechanical stability, a test was conducted by recording the output voltage of the TENGs fabricated with PVDF-500 and PVDF/BTO@PDA-500 over 100 000 cycles at a force and frequency of 30 N and 2 Hz, respectively. Figure 8g shows the retention of output voltages for the PVDF-500 and PVDF/BTO@PDA-500 throughout the 100 000 cycles. Initially, the output voltage exhibited a gradual increase for both PVDF-500 and PVDF/BTO@PDA-500. This can be attributed to the porous structure of NF mat. The charges in the surface layer transferred to the inner layer of the NF mat during the compression of the pores, thereby accumulating within the mat as the operating cycle progressed [72]. However, the output voltage of PVDF-500 decreased after reaching 40 000 cycles, whereas that of PVDF/BTO@PDA-500 remained

stable with no considerable decrease over 100 000 cycles. The enhanced stability of the PVDF/BTO@PDA NF mat can be attributed to the incorporation of nanofillers within the polymer matrix, which are known to improve the mechanical properties [73]. The voltage signals of PVDF-500 and PVDF/BTO@PDA-500 at every 20 000 cycles are shown in Fig. 8h and i, respectively. Additionally, the patterned PVDF/BTO@PDA NF mats maintained their original morphologies even after applying a considerable force, regardless of the mesh number (Fig. S10). These results reveal the durability of the patterned PVDF/BTO@PDA NF mats.

4 Conclusion

In this study, patterned PVDF/BTO@PDA NF mats were fabricated using a metal mesh as the NF collector for application in TENGs. After detaching the NF mat from the mesh collector, the surface of the NF mat that was in contact with the mesh was used as the contact layer of the negative triboelectric material to preserve its unique pattern from the extended electrospinning time. The output voltage and the current of the TENG were considerably enhanced using the patterned PVDF NF mat as a negative triboelectric material owing to the increased effective contact area between the two contact layers. The TENG with PVDF-500 achieved the highest voltage owing to the high density of the NF patterns. Regarding the NF mats, the output voltage remained unsaturated with the applied force of 0–30 N, thereby indicating that the protruding NF patterns did not undergo complete deformation. The performance of the TENG was further enhanced by inducing a synergetic effect of nonadditive and additive physical modification methods for the PVDF NF mat. Notably, the crystallinity of PVDF and dispersion of BTO NPs considerably improved when the NP surface was functionalized using PDA. Consequently, PVDF/BTO@PDA-500 exhibited the highest voltage output, which is attributed to its increased β-phase content and enhanced surface potential measured using KPFM. Utilizing large BTO NPs increased the surface roughness of the NFs, thereby resulting in a nanopatterning effect. Consequently, the highest TENG performance was obtained when PVDF/BTO@PDA-500 with 400-nm NPs was used as the negative triboelectric material. Further, the origin of the voltage signal of the TENG was elucidated by analyzing the frequency domain signals. Voltage signals were generated owing to the piezoelectric effect of the nanocomposite PVDF/BTO, triboelectric effect between the two contact layers, and internal contact between the PVDF NFs and Al electrode.

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Data Availability Data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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